
When Adapter Training Hurts: Manifold Preservation for Vector Retrieval

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Abstract

Adapting pre-trained vision-language models for vector retrieval typically relies on global contrastive objectives, which optimize semantic alignment across all training categories simultaneously. We identify a fundamental limitation of this approach: when applied through high-capacity adapters, global objectives produce dense gradient fields that systematically distort the pre-trained manifold of unseen categories, causing out-of-distribution (OOD) Label Precision to collapse below the frozen baseline. Although parameter-efficient low-rank adapters such as LoRA largely preserve performance when trained with global losses like InfoNCE or triplet objectives due to their capacity bottleneck, they do not exploit local sparse updates to the same extent. To address this, we propose Euclidean Geodesic Alignment (EGA), a residual adapter trained with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings. By operating locally rather than globally, EGA converges to a sparse-gradient state in which 96.5% of training triplets produce zero gradient at steady state, leaving the broader embedding space, including regions occupied by unseen categories, structurally intact. Extensive evaluations on CIFAR-10, CIFAR-100, FGVC-Aircraft, and Food-101 demonstrate that EGA consistently outperforms global contrastive methods including ICon and SRL across both in-distribution and OOD settings while also surpassing or matching the LoRA+InfoNCE and LoRA+Triplet baselines. On strictly unseen categories, EGA improves Label Precision over the frozen baseline on Aircraft and CIFAR-10 and bounds the degradation on Food-101, while ICon and SRL fall below the frozen baseline by up to 36 percentage points.

1 Introduction

Modern retrieval systems increasingly rely on embedding models such as CLIP [24] for their remarkable zero-shot semantic capabilities. These models provide highly generalizable latent spaces that serve as excellent starting points for vector similarity search. However, when deploying these systems to specialized domains, practitioners often seek to further align the embeddings with domain-specific category structures to maximize retrieval accuracy. Because full scale fine-tuning is computationally prohibitive and risks destroying the foundational zero-shot knowledge, parameter-efficient adaptation has emerged as the standard paradigm. In this approach, lightweight adapters are trained on a labeled subset of the target task while the massive pre-trained backbone remains frozen.

Despite the elegance of this parameter-efficient paradigm, the training of these adapters is almost universally driven by global contrastive objectives. These methods compute losses over the full pairwise similarity matrix in each mini batch, continuously reshaping the entire embedding distribution to enforce global alignment and uniformity. While these objectives successfully boost retrieval accuracy for in-distribution queries where all relevant categories are present during training, they introduce a critical failure mode for real-world retrieval systems that must inevitably handle unseen categories.

We reveal that when paired with high-capacity adapters, global contrastive methods such as ICon [1] and SRL [4] systematically distort the pre-trained semantic manifold of unseen classes. Although the resulting embeddings appear geometrically optimized for IVF-based indexing, they frequently retrieve neighbors of the wrong semantic class. This distortion causes out-of-distribution Label Precision to drop by 4 to 36 percentage points below the raw frozen baseline across multiple benchmarks. In contrast, while low-rank adapters such as LoRA [10] largely avoid this catastrophic OOD degradation, their restricted capacity inherently limits their ability to fully capture complex semantic alignments on seen domains. Consequently, practitioners face a fundamental dilemma: scaling adapter capacity under global objectives trades semantic correctness for geometric indexability, whereas restricting capacity compromises overall retrieval efficacy.

To resolve this fundamental dilemma, we propose Euclidean Geodesic Alignment (EGA), a manifold-preserving adapter that breaks the tradeoff between model capacity and semantic integrity by replacing dense global optimization with sparse local updates. EGA incorporates three synergistic design choices: (i) a zero-initialized residual structure, (ii) an ℓ_2 normalization constraint, and (iii) a small-margin triplet loss operating on a class-aware neighborhood graph. Together, these components restrict the adapter’s influence to local neighborhood relationships that explicitly violate the target retrieval objective. As training proceeds, the fraction of triplets producing a non-zero gradient decays rapidly. At convergence, 96.5% of the training triplets contribute zero gradient. Consequently, the high-capacity adapter ceases to modify the vast majority of the embedding space, successfully preserving the pre-trained semantic structure for unseen categories and achieving bounded degradation even under heavy out-of-distribution shifts.

We evaluate EGA across in-distribution (CIFAR-100) and three out-of-distribution benchmarks (CIFAR-10, FGVC-Aircraft, Food-101). We compare EGA against the frozen CLIP baseline, global contrastive adapters (Icon and SRL), and low-rank adapters (LoRA). On in-distribution data, EGA consistently improves retrieval quality. On unseen categories, EGA improves Label Precision over the frozen baseline on Aircraft and CIFAR-10 and bounds degradation on Food-101. In contrast, ICon and SRL fall below the frozen baseline on every out-of-distribution dataset, dropping by 32 to 36 percentage points on Food-101. While LoRA avoids this collapse, it underperforms EGA due to capacity constraints. On FGVC-Aircraft, EGA achieves a Label Precision of 0.548, outperforming the frozen baseline (0.512), ICon (0.423), and SRL (0.441).

Contributions This paper makes the following contributions: (i) We identify that dense optimization gradients from global contrastive adapters distort the pre-trained manifold of unseen categories; (ii) We propose Euclidean Geodesic Alignment (EGA) to refine metric geometry via a zero-initialized residual adapter and local triplet loss; and (iii) We demonstrate that EGA consistently improves out-of-distribution Label Precision over the frozen baseline, outperforming state-of-the-art methods by a margin of up to 36 percentage points (pp).

2 Related Work

Manifold Preservation under Adapter Tuning Foundation models such as CLIP [24] produce embedding spaces whose structure encodes both seen class semantics and a broader prior over visual concepts not observed during downstream training. Catastrophic forgetting has been studied extensively in continual and incremental learning settings [20, 36]. Recent work has documented analogous degradation phenomena specific to contrastive representations: anisotropic collapse into narrow conic regions [41, 6], structural fracturing under modality gap [25, 5], and loss of compositional structure under aggressive optimization [23, 27, 35]. To counter these degradations, recent adaptation strategies impose explicit geometric constraints such as orthogonality regularization [26] or restrict optimization to limited subspaces. EGA leverages the inherent sparsity of triplet loss optimization to confine adapter updates to local neighborhoods of seen classes, leaving the unseen class regions of the pretrained manifold structurally unmodified.

Global Contrastive Objectives and Structural Regularization Recent advancements in representation learning often rely on global contrastive objectives to optimize the latent space [30, 19]. A common paradigm involves jointly enforcing alignment for positive pairs and uniformity for the overall feature distribution [33], a strategy applied across various domains including graph encoders [32] and multimodal recommendation systems [42]. While pushing embeddings toward a uniform distribu-

tion enhances robustness to background noise [34], it inherently assumes that all latent regions require the same degree of global scattering. To further refine complex distributions, structural regularization techniques introduce auxiliary constraints. For instance, methods like ICon formulate a unifying framework to systematically align multi view representations [1], while geometric constraints are leveraged to improve imbalanced regression tasks [4]. Other approaches attempt to decouple the learning objectives [18] or apply information theoretic regularizers [40] to control the density of the learned representations. Furthermore, region based distillation techniques have been proposed to enhance fine grained understanding [15]. EGA replaces dense global optimization with sparse local triplet updates, preserving out of distribution semantic structure.

Parameter-Efficient Manifold Adaptation Parameter efficient adaptation emerged as a standard to prevent catastrophic forgetting and reduce computational overhead when transferring large foundation models [9, 10]. This paradigm has extensively shifted towards vision language models, dominating recent literature due to the prohibitive cost of full scale fine tuning [21]. Methods such as Visual Prompt Tuning [14] and Tip Adapter demonstrate that lightweight tuning can effectively specialize representations for downstream tasks without altering the frozen backbone [39]. Furthermore, extensive benchmarking confirms that parameter efficient techniques can rival full fine tuning while preserving the generalizable representations of the foundation model [28]. To further protect the pre trained manifold during initial training steps, zero initialization strategies have proven highly effective. For instance, conditional control mechanisms in image generation initialize adapter weights to zero, ensuring the model behaves exactly like the frozen base model before any optimization occurs [38]. EGA integrates a zero initialized residual adapter with a localized geometric objective, achieving both parameter efficiency and targeted manifold smoothing.

Vector Similarity Search and Metric Consistency Vector similarity search is the foundational engine for retrieving high dimensional representations at scale. The systems community has developed highly optimized indexing algorithms to handle billion scale datasets. Foundational works such as FAISS leverage hardware acceleration for fast similarity search [16], and Hierarchical Navigable Small World graphs establish the core topology for efficient approximate retrieval [22]. Meanwhile, hybrid structures like DiskANN and SPANN achieve exceptional speed by efficiently managing memory and disk hierarchies [29, 2]. Recent advancements continue to refine these topologies, exploring crossing sparse proximity graphs [37] and analyzing the theoretical constructions of navigable graphs for high dimensional nearest neighbor search [3]. Furthermore, system level studies optimize retrieval latency and stability by aligning best first search algorithms with solid state drives [7] and enabling direct insertions in large scale indices [8]. To reduce the computational cost of exact distance calculations during the retrieval phase, approximation techniques such as low rank matrix factorization have also been introduced to accelerate inner product evaluations [13]. However, extensive theoretical analyses reveal that popular approximate nearest neighbor implementations still face severe worst case performance limitations [12]. EGA proactively recalibrates the metric manifold prior to index construction, enabling standard search backends to achieve superior semantic recall without requiring complex algorithmic reengineering.

3 Methodology

3.1 Problem Formulation

Let $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$ denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image $x \in \mathcal{X}$ to a unit-normalized embedding $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$. We are given a set of labeled training images $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^N$, where $y_i \in \mathcal{Y}_{\text{seen}}$ belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes $\mathcal{Y}_{\text{unseen}}$, with $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$.

Our goal is to learn a lightweight adapter $f_\theta : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$ such that the transformed embeddings $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to any unseen-class labels during training.

Evaluation metrics. We evaluate retrieval quality along two complementary dimensions.

Label Precision@ K measures the semantic quality of retrieved results—whether the K nearest neighbors in the transformed space share the same class label as the query:

$$\text{LP@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^K \mathbf{1}[y_{\hat{n}_k(q)} = y_q], \quad (1)$$

where $\mathcal{Q}$ is the query set and $\hat{n}_k(q)$ denotes the k -th nearest neighbor of q in the transformed embedding space.

ANNS Recall@ K measures the geometric fidelity of the ANN index—the fraction of true K nearest neighbors (under exact search) that are recovered by the approximate index at a given search budget n_{probe} :

$$\text{AR@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)|}{K}, \quad (2)$$

where $\mathcal{N}_K^{\text{exact}}(q)$ and $\mathcal{N}_K^{\text{approx}}(q)$ denote the exact and approximate top- K neighbor sets, respectively.

The two metrics capture distinct failure modes: an adapter may improve LP@ K by tightening within-class clusters (semantic improvement) while simultaneously improving AR@ K by making the geometry more amenable to IVF-based indexing (geometric improvement). As we show in Section 4, global contrastive objectives achieve high AR@ K on seen classes but collapse on *both* metrics for unseen classes, whereas EGA improves both simultaneously under the OOD setting.

3.2 Manifold-Preserving Adapter Design

EGA introduces a lightweight adapter f_θ composed of stacked residual blocks with a final refinement layer. The architecture is deliberately constrained by three design principles, each of which serves a distinct role in preserving the pretrained manifold.

Residual connection with zero initialization. Each EGA block applies a residual update of the form

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \quad (3)$$

where g_θ is a learned transformation initialized so that $g_\theta(\mathbf{z}) \approx \mathbf{0}$ at the start of training. Concretely, the final linear layer of g_θ is initialized to zero, making f_θ an identity mapping at initialization. This guarantees that training begins from the pretrained geometry rather than a random perturbation, and that the adapter can only *refine* rather than replace the CLIP embedding. As confirmed by our ablation (Table 2), removing the residual connection causes accuracy to collapse from 0.70 to 0.01, because the lightweight MLP cannot reconstruct complex semantic relationships from seen-class data alone.

Feature gating. Inside each residual block, the transformed signal passes through a feature-gating module:

$$\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2 \text{GELU}(W_1 \mathbf{h})), \quad (4)$$

where $W_1 \in \mathbb{R}^{(d/4) \times d}$, $W_2 \in \mathbb{R}^{d \times (d/4)}$, σ is the sigmoid function, and $\odot$ denotes elementwise multiplication. This structure, analogous to Squeeze-and-Excitation networks [11], learns a per-dimension importance weight that amplifies retrieval-relevant dimensions and suppresses noisy ones. The bottleneck ratio of 1/4 is chosen to prevent the gating network from overfitting on seen-class data, as validated in Table 3.

ℓ_2 normalization. The final output of f_θ is projected back onto the unit hypersphere $\mathbb{S}^{d-1}$ via ℓ_2 normalization:

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \quad (5)$$

This constraint ensures that the transformed embeddings remain in the same metric space as the original CLIP embeddings, so that Euclidean distance and cosine similarity remain interchangeable on the hypersphere. Removing this constraint degrades retrieval performance by 4.2 percentage points (Table 2), confirming that hypersphere consistency is necessary for stable ANN indexing.

179 **Full forward pass.** Let Block_i denote the i -th EGA block consisting of a two-layer MLP with LayerNorm followed by feature gating, and let Refine denote the final linear projection with LayerNorm. 180 The complete forward pass for an input embedding $\mathbf{z} \in \mathbb{S}^{d-1}$ is: 181

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z})))) . \quad (6)$$

182 The total number of trainable parameters is approximately $4.2M$ for $d = 512$ and hidden dimension 183 2048, which is negligible compared to the frozen CLIP backbone ($\sim 150M$ parameters).

184 3.3 Local Triplet Training and Gradient Sparsity

185 **Training objective.** Given a mini-batch sampled from $\mathcal{D}_{\text{train}}$, we construct triplets (a, p, n) where a 186 is an anchor image, p is a positive sampled uniformly from the same class as a , and n is a negative 187 sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \max(0, d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m), \quad (7)$$

188 where $d(\cdot, \cdot)$ denotes Euclidean distance, $m > 0$ is the margin hyperparameter, and $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$ is the 189 transformed embedding. We set $m = 0.2$ throughout all experiments based on the sensitivity analysis 190 in Table 3.

191 **Gradient sparsity and manifold preservation.** A triplet (a, p, n) contributes a non-zero gradient 192 to θ if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \quad (8)$$

193 We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many 194 triplets satisfy the margin constraint and become inactive, contributing zero gradient. Figure 5 (left) 195 shows that the active triplet ratio decays from 17.4% at epoch 10 to 3.5% at convergence. At steady 196 state, **96.5% of triplets produce zero gradient**, meaning the adapter ceases to modify the vast 197 majority of local neighborhood relationships in the embedding space.

198 This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes 199 are absent from $\mathcal{D}_{\text{train}}$, no triplet involving an unseen-class embedding is ever constructed. Provided 200 that the adapter’s updates remain local (i.e., that gradients from seen-class triplets do not propagate 201 into unseen-class regions of the embedding space), the CLIP manifold structure for unseen classes 202 is preserved intact. The gradient sparsity mechanism enforces exactly this locality: once seen-class 203 neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of 204 actively-violated local relationships continue to be updated.

205 **Contrast with global contrastive objectives.** Global objectives such as ICon [1] and SRL [4] 206 compute losses over the *full* pairwise similarity matrix within each mini-batch. ICon minimizes a 207 KL divergence between the empirical similarity distribution and the class-membership distribution, 208 while SRL jointly optimizes batch-wide uniformity and within-class homogeneity. Both objectives 209 generate dense gradient fields in every training step: every sample pair (regardless of whether its 210 local geometry already satisfies the retrieval objective) contributes to the loss and receives a non-zero 211 gradient. This global optimization pressure from seen-class training propagates throughout the 212 embedding space, reshaping the relative positions of unseen-class regions according to seen-class 213 similarities rather than preserving the pretrained semantic structure for those categories.

214 **Proposition 1** (Gradient Sparsity at Convergence). *Let f_θ be trained with the triplet loss in Eq. (7) 215 with margin $m > 0$ on a dataset $\mathcal{D}_{\text{train}}$ drawn from seen classes $\mathcal{Y}_{\text{seen}}$. Define the active triplet ratio 216 at training step t as*

$$\rho(t) = \Pr_{(a,p,n) \sim \mathcal{B}} \left[d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}) - d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}) + m > 0 \right]. \quad (9)$$

217 *As f_θ converges, $\rho(t) \rightarrow \rho^*$, where*

$$\rho^* = \Pr_{(a,p,n)} \left[d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m > 0 \right] \quad (10)$$

218 *is determined solely by the margin m and the within-class distance distribution of the converged 219 embeddings. Furthermore, $\nabla_\theta \mathcal{L} = \mathbf{0}$ for all triplets (a, p, n) with $d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m \leq 0$, so 220 the adapter gradient is supported only on the fraction ρ^* of the training distribution. For embeddings 221 whose local geometry already satisfies the margin constraint, the adapter exerts zero update pressure.*

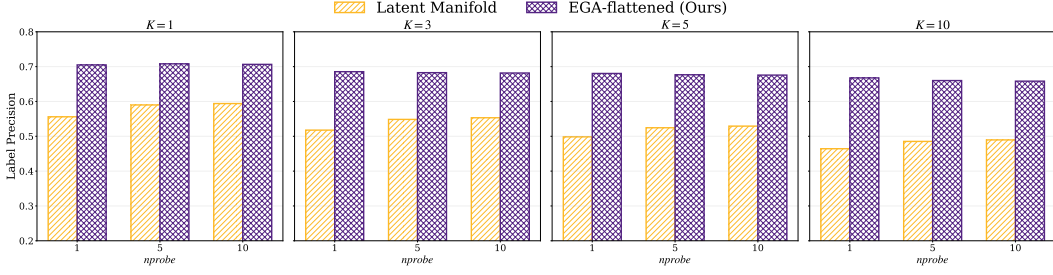


Figure 1: Label Precision@ K on CIFAR-100 across $K \in \{1, 3, 5, 10\}$ and $nprobe \in \{1, 5, 10\}$. EGA consistently improves over the raw CLIP latent space.

Proof Sketch. The triplet loss consists of hinge terms whose gradient is zero whenever the margin constraint is satisfied. At convergence, only those triplets whose geometry the adapter has not yet separated within the margin remain active. The fraction of such triplets is determined solely by the margin m and the distance distribution of seen classes, independent of unseen classes. Since no training triplet involves unseen-class embeddings, the adapter exerts zero gradient over the entire unseen-class region of the embedding space. See Appendix A for the complete proof. $\square$

4 Evaluation

We answer the following research questions through extensive experiments:

- RQ1: Does EGA improve retrieval quality for in-distribution queries, and how does it affect the geometric structure of the embedding space? (Section 4.2)
- RQ2: Under strict out-of-distribution settings, how does EGA perform on both Label Precision and ANNS Recall compared to global contrastive methods and low-rank adapters? (Section 4.3)
- RQ3: What is the underlying mechanism that enables triplet trained adapters to preserve unseen class structures while global contrastive objectives destroy them? (Section 4.4)
- RQ4: How do the individual design choices and hyperparameters impact the overall retrieval effectiveness and stability of EGA? (Section 4.5)

4.1 Experimental Setup

Experiments were conducted on the Chameleon Cloud platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. Evaluation datasets include CIFAR-100 for in distribution scenarios, CIFAR-10 for cross dataset OOD settings, and FGVC-Aircraft and Food-101 for unseen class OOD evaluations. All of these datasets can be directly downloaded from the Pytorch library. We compare our method against four baseline models: ICon [1], SRL [4], LoRA + InfoNCE [10, 31], and LoRA + Triplet [10].

Extended experimental details can be found in Appendix B.

4.2 In-Distribution Validation

Semantic retrieval quality. Figure 1 reports Label Precision@ K across four neighborhood sizes and three search budgets. EGA improves Label Precision@1 from ≈ 0.55 (raw CLIP) to > 0.70 at $nprobe = 1$, and the gap persists across all K and $nprobe$ values. The improvement is not an artifact of larger search budgets: even at $nprobe = 10$, EGA retains a clear advantage, indicating that the adapter genuinely tightens within-class neighborhoods rather than merely benefiting from broader index coverage.

Geometric structure. The Label Precision improvement has a direct geometric correlate. Figure 2 contrasts the L_2 distance distributions for true Top- K neighbors against random background points, before and after EGA. In the raw CLIP space, the two distributions overlap substantially: the distance

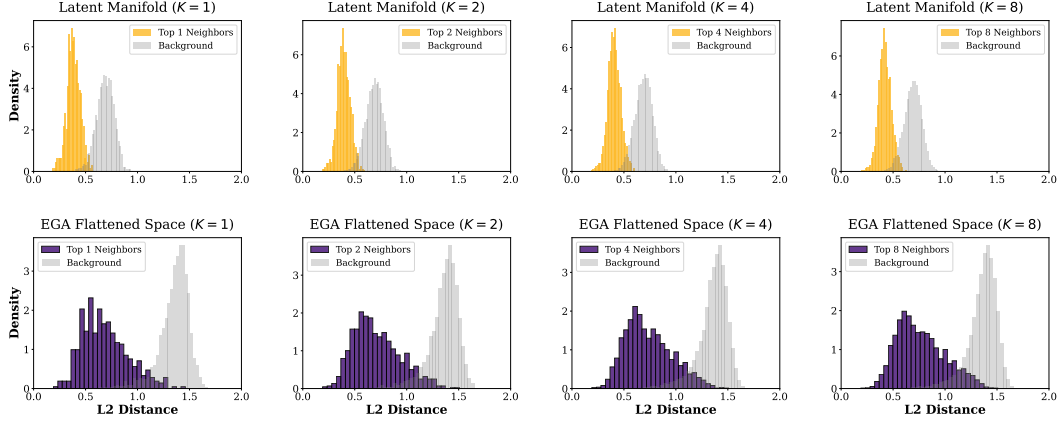


Figure 2: L_2 distance distributions for Top- K neighbors versus random background points on CIFAR-100. The raw CLIP space exhibits significant overlap; EGA separates the two distributions cleanly.

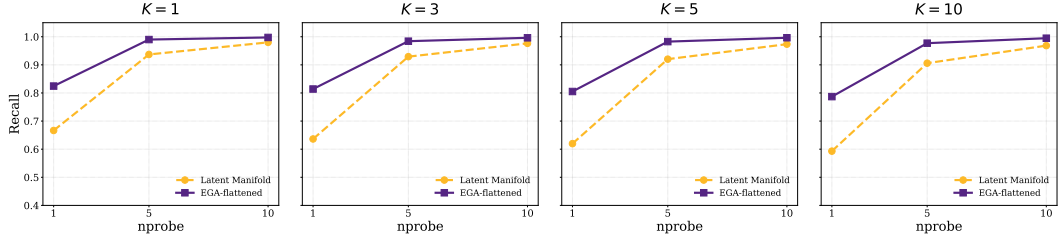


Figure 3: ANNS Recall@ K versus nprobe on CIFAR-100. EGA shifts the efficiency frontier upward across all K .

to a true neighbor is often indistinguishable from the distance to an unrelated point, which is exactly the condition under which IVF-based ANN indexing degrades. After EGA, the two distributions separate cleanly across all K , with neighbors concentrated at small distances and background points pushed toward a distinct mode. This separation is the mechanism by which EGA improves the recall–efficiency frontier shown in Figure 3: the raw CLIP space achieves only 0.67 Recall@1 at nprobe = 1, whereas EGA reaches 0.80 and saturates above 0.97 by nprobe = 5.

4.3 Out-of-Distribution Generalization

We now turn to the central empirical claim of this paper. Under strict OOD settings where evaluation categories are disjoint from training categories, the two retrieval metrics tell different stories about adapter quality, and EGA is the only method that performs well on both. Table 1 reports headline numbers at $K = 1$, nprobe = 1, and Figure 4 traces Label Precision across nprobe $\in \{1, 5, 10\}$ to show that the pattern is robust to search budget. We evaluate on three benchmarks: CIFAR-10 (trained on CIFAR-100), FGVC-Aircraft (80 seen / 20 unseen classes), and Food-101 (80 seen / 21 unseen classes). For a fair comparison, all methods train identical lightweight adapters on top of frozen CLIP features and differ only in their loss functions.

On ANNS Recall, every adapter improves or matches the frozen CLIP baseline across all three OOD benchmarks (Table 1). Adapter training succeeds at *geometric* reorganization: the resulting embedding distribution is more amenable to IVF-based indexing. On Label Precision, however, the picture inverts. ICon and SRL fall *below* the frozen CLIP baseline on every dataset, with degradation reaching -32 to -36 percentage points on Food-101. The retrieved neighbors are geometrically closer in the IVF index, but they belong to the wrong semantic class. EGA is the only adapter that improves Label Precision on Aircraft (+3.6 pp) and CIFAR-10 (+0.6 pp), and on Food-101 it retains the strongest Label Precision among all adapters with bounded -8.3 pp degradation.

The Label Precision degradation of global contrastive methods persists across all three datasets and all three search budgets we tested (18 measurement points in total, Figure 4), ruling out the possibility

Table 1: OOD performance at $K = 1$, $nprobe = 1$.

Dataset	Metric	CLIP	ICoN	SRL	LoRA+InfoNCE	LoRA+Triplet	EGA
CIFAR-10	AR@1	0.863	0.913	0.942	0.856	0.866	0.986
	LP@1	0.880	0.841	0.794	0.880	0.870	0.886
Aircraft	AR@1	0.773	0.875	0.863	0.911	0.881	0.898
	LP@1	0.512	0.423	0.441	0.560	0.577	0.548
Food-101	AR@1	0.842	0.811	0.827	0.902	0.899	0.886
	LP@1	0.881	0.557	0.520	0.878	0.833	0.798

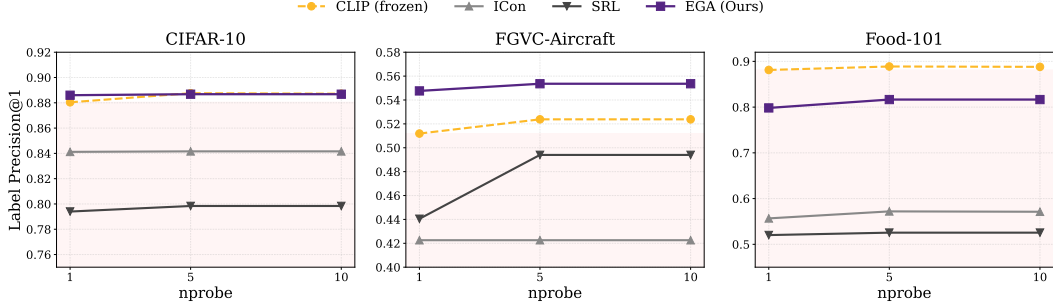


Figure 4: Label Precision@1 across three OOD benchmarks and three search budgets.

that it is an artifact of a particular benchmark or indexing configuration. EGA’s relative advantage over global methods ranges from 4.5 pp on CIFAR-10 to 24 pp on Food-101, and tracks the difficulty of the OOD shift: when seen and unseen classes are visually similar (as in Food-101’s fine-grained dishes), the dense gradient updates of ICoN and SRL pull unseen samples most aggressively into the seen seen-class clusters.

We additionally evaluate LoRA [10] as a parameter-efficient baseline (rank $r = 8$, $\sim 8K$ trainable parameters versus $\sim 4.2M$ for the EGAMLP architecture used by EGA, ICoN, and SRL). Two configurations are included: LoRA + InfoNCE (a global contrastive loss) and LoRA + Triplet (the same loss as EGA). Both LoRA configurations avoid the catastrophic Label Precision collapse exhibited by ICoN and SRL even when paired with a global loss: LoRA + InfoNCE on Food-101 reaches LP 0.878, only 0.3 pp below the frozen baseline. This suggests that the catastrophic failure of ICoN and SRL arises from the *combination* of (i) full-capacity adapter and (ii) dense global gradients, not from the global loss alone. With an aggressive bottleneck on adapter capacity, global losses become survivable.

4.4 Gradient Sparsity as OOD Protection

The OOD pattern in Section 4.3 raises a mechanistic question: why does EGA’s triplet objective preserve unseen-class structure while global contrastive objectives reshape it? We identify gradient sparsity at convergence as the key mechanism.

A triplet loss with margin m produces a non-zero gradient for a sample triplet (a, p, n) only when the margin constraint is violated, i.e., when

$$d(a, p) - d(a, n) + m > 0. \quad (11)$$

As the adapter converges on seen classes, the fraction of such *active* triplets decays rapidly. Figure 5 (left) tracks this ratio throughout training: starting at 17.4% at epoch 10, it falls to 3.5% by convergence. At steady state, 96.5% of triplets contribute zero gradient, meaning the adapter updates only a sparse subset of local neighborhood relationships and leaves the remainder of the embedding space structurally intact.

Global contrastive objectives such as ICoN [1] and SRL [4] exhibit the opposite behavior. ICoN minimizes a KL divergence over the full pairwise similarity matrix, and SRL jointly optimizes batch-wide uniformity and homogeneity. Both produce *dense* gradient fields in every training step: every sample pair contributes to the loss regardless of whether its local geometry already satisfies the retrieval objective. This global optimization pressure shapes the entire embedding distribution based

Table 2: Ablation study of EGA

Variant	Accuracy
Full EGA (Ours)	0.7015
w/o Residual connection	0.0065
w/o Zero-initialization	0.6840
w/o L2-normalization	0.6590

Table 3: Sensitivity of different hyperparameters

Triplet Margin m	0.1	0.2	0.3	0.5
Recall@1	0.8420	0.8180	0.8160	0.7400
Recall@5	0.9782	0.9654	0.9610	0.9125
Hidden Dimension d	512	1024	2048	4096
Recall@1	0.8300	0.8340	0.8200	0.8240
Recall@5	0.9695	0.9712	0.9680	0.9688

on seen-class similarities, even for regions of the space occupied by unseen-class samples. The result is not loss of geometric structure but rather loss of *semantically correct* structure: unseen samples are pulled toward the clusters of seen classes that they happen to resemble in CLIP’s original metric, rather than toward other unseen samples that share their true category.

Figure 5 (right) and Figure 4 together quantify the consequence: across all three OOD benchmarks, ICon and SRL fall below the frozen CLIP baseline on Label Precision (e.g., on Aircraft, ICon collapses to 0.423 from a baseline of 0.512), confirming that dense gradient updates actively harm zero-shot semantic retrieval. EGA, by contrast, improves Label Precision on Aircraft and CIFAR-10 while only 3.5% of training triplets contribute non-zero gradients at convergence.

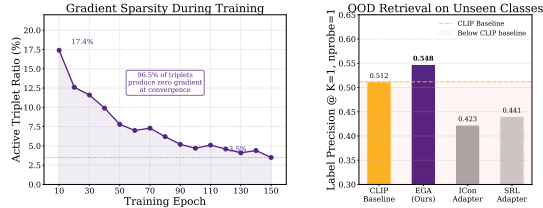


Figure 5: Gradient sparsity as OOD protection.

4.5 Ablation and Sensitivity Analysis

Table 2 isolates the contribution of each architectural component. The residual connection is the core element. Removing it causes accuracy to collapse from 0.7015 to 0.0065. This collapse indicates that the adapter cannot reconstruct semantic relationships from target data alone and must operate as a refinement over the pre-trained features. The L2 normalization ensures the transformed embeddings remain on the unit hypersphere. Removing this constraint decreases accuracy to 0.6590. Zero-initialization allows the adapter to begin training as an identity mapping. Removing it lowers accuracy to 0.6840.

Table 3 evaluates model stability across hyperparameter variations. EGA maintains consistent retrieval quality across margin values from 0.1 to 0.3. Increasing the margin to 0.5 degrades Recall@1 to 0.7400. A larger margin forces the adapter to update embeddings that already satisfy local neighborhood constraints, which reduces gradient sparsity and alters the pre-trained manifold. Furthermore, the model shows stable performance across hidden dimensions ranging from 512 to 4096. This stability confirms that EGA achieves metric correction through its structural design rather than specific parameter capacities.

5 Conclusion

High-capacity adapters trained with global contrastive objectives harm out-of-distribution *semantic* retrieval, even when they improve geometric indexability. Under these objectives, Label Precision on unseen categories falls below the frozen baseline. EGA resolves this failure mode by converging to a sparse gradient strategy. At steady state, 96.5% of training triplets contribute zero gradient. This sparsity preserves the pre-trained semantic structure for unseen categories. Across three out-of-distribution benchmarks (CIFAR-10, FGVC-Aircraft, Food-101), EGA improves Label Precision over the frozen baseline on Aircraft and CIFAR-10, and bounds the degradation on Food-101. In contrast, ICon and SRL degrade by up to 36 percentage points. These results establish gradient sparsity as a design principle for retrieval adapters operating under distribution shift.

Limitations. This study focuses on visual representations from CLIP. Future work will extend EGA to text and multimodal models, explore unsupervised variants, and disentangle the interaction between adapter capacity and loss sparsity.

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A Proof of Proposition 1

Proof. We prove each claim in turn.

(1) Gradient support. The triplet loss in Eq. (7) can be written as $\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \ell_{apn}$, where $\ell_{apn} = \max(0, \delta_{apn} + m)$ and $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$. The subgradient with respect to θ is

$$\nabla_{\theta} \ell_{apn} = \mathbf{1}[\delta_{apn} + m > 0] \cdot \nabla_{\theta} \delta_{apn}, \quad (12)$$

which is exactly zero whenever $\delta_{apn} + m \leq 0$. Therefore, $\nabla_{\theta} \mathcal{L}$ receives contributions only from active triplets satisfying Eq. (8).

(2) Convergence of $\rho(t)$. Under mild regularity conditions (Lipschitz continuity of f_{θ} and bounded embedding norms), the sequence $\{\hat{\mathbf{z}}^{(t)}\}$ converges to a stationary point $\hat{\mathbf{z}}^*$ of $\mathcal{L}$. At stationarity, $\rho(t)$ converges to ρ^* as defined in Eq. (10), since $\rho(t)$ is a continuous functional of the embedding distribution and the embeddings converge.

(3) Zero gradient over unseen regions. Let $\mathcal{R}_{\text{unseen}} \subset \mathbb{S}^{d-1}$ denote the region of the embedding space occupied by unseen-class images under f_{θ} . By construction, all triplets (a, p, n) in $\mathcal{D}_{\text{train}}$ satisfy $y_a, y_p, y_n \in \mathcal{Y}_{\text{seen}}$, so $\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p, \hat{\mathbf{z}}_n \notin \mathcal{R}_{\text{unseen}}$ for all training triplets. The gradient $\nabla_{\theta} \delta_{apn}$ depends on f_{θ} only through its evaluation at $\mathbf{z}_a, \mathbf{z}_p, \mathbf{z}_n$, which lie outside $\mathcal{R}_{\text{unseen}}$. Consequently, no training step produces a gradient that directly modifies f_{θ} in $\mathcal{R}_{\text{unseen}}$, and the CLIP geometry in that region is preserved throughout training.

(4) Dependence on m . The residual distance gap δ_{apn}^* at convergence is bounded by the within-class diameter $\Delta_c = \max_{x, x' \in c} d(\hat{\mathbf{z}}, \hat{\mathbf{z}}')$ for seen class c . For $m \leq \min_c \Delta_c$, the adapter can drive all within-class distances below the threshold, pushing ρ^* toward zero. For $m > \min_c \Delta_c$, some triplets remain active at convergence because the within-class spread exceeds the margin, resulting in a strictly positive ρ^* . This explains the empirically observed increase in ρ^* with larger m reported in Table 3. $\square$

B More Experimental Details

Datasets. We evaluate on four datasets covering both in-distribution and OOD retrieval scenarios. **CIFAR-100** (100 classes, 60K images) serves as the in-distribution benchmark: the adapter is trained and evaluated on the same class label space, with a 75/25 split between database and query sets. **CIFAR-10** (10 classes, 60K images) provides a cross-dataset OOD setting: the adapter is trained on CIFAR-100 and evaluated on the disjoint CIFAR-10 classes. **FGVC-Aircraft** (100 fine-grained aircraft variants, 10K images) and **Food-101** (101 food categories, 25K images for the test split we use) provide unseen-class OOD settings: classes are randomly partitioned into 80% seen / 20% unseen with a fixed seed, the adapter is trained on the seen split, and retrieval is evaluated strictly on the unseen split. Image embeddings for all datasets are extracted with frozen CLIP ViT-B/32.

Baselines. We compare EGA against the frozen CLIP baseline and four trained adapters, all sharing the same training data, evaluation protocol, and 75/25 retrieval split:

- **Icon** [1]: a global contrastive objective minimizing the KL divergence between the empirical similarity distribution and the class-membership distribution. We instantiate Icon on the same EGAMLP architecture used by our method (~ 4.2 M parameters).
- **SRL** [4]: a structural regularization objective jointly optimizing batch-wide uniformity and within-class homogeneity, also instantiated on the EGAMLP architecture.
- **LoRA + InfoNCE** [10, 31]: a low-rank adapter (rank $r = 8$, $\alpha = 16$, ~ 8 K parameters) with zero-initialized residual structure, trained with supervised InfoNCE, which is a standard global contrastive loss.
- **LoRA + Triplet** [10]: identical low-rank architecture as above, but trained with the same small-margin ($m = 0.2$) triplet loss as EGA.

The two LoRA configurations isolate the effect of architectural capacity vs. loss function: comparing LoRA + InfoNCE against Icon/SRL holds the loss family roughly fixed while changing capacity, and comparing LoRA + Triplet against EGA holds the loss fixed while changing capacity.

873 **Indexing and metrics.** For all retrieval evaluations we use FAISS [16] IndexIVFFlat with
874 nlist=10 centroids and report results at $nprobe \in \{1, 5, 10\}$. We measure both Label Preci-
875 sion (LP@ K , Eq. 1) and ANNS Recall (AR@ K , Eq. 2) at $K \in \{1, 3, 5, 10\}$.

876 **Training.** All adapters are trained for 150 epochs with AdamW (weight decay 10^{-4}), cosine-
877 annealed learning rate, and batch size 256. EGAMLP-based methods (EGA, ICon, SRL) use learning
878 rate 10^{-4} ; LoRA variants use 10^{-3} to compensate for their smaller parameter count. Triplet-based
879 methods sample one positive and one negative per anchor uniformly within each mini-batch; global
880 contrastive methods (ICoN, SRL, LoRA + InfoNCE) operate on the full pairwise similarity matrix.
881 All experiments use a fixed random seed (42) for reproducibility.

882 **Compute platform.** All experiments were conducted on the Chameleon Cloud [17] platform using
883 a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA
884 A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS.